

Multivariate practical 7

PCA

Practical

Introduction

Survey data to assess options for enhancing food security through sustainable use of natural resources in the Sikasso region of Mali is available in `data_tidy_mali` (instructions to get it below). Use R to analyse the data and interpret your results. Use comments to show what you are doing and what it all means.

Variables include:

- family (total number of individuals in household)
- distrd (distance in kilometres to nearest passable road)
- cotton (hectares of cotton planted in year 2000)
- maize (hectares of maize planted in year 2000)
- sorg (hectares of sorg planted in year 2000)
- millet (hectares of millet planted in year 2000)
- bull (total number of bullocks or draft animals)
- cattle (total)
- goats (total)

Questions

Download and attach the dataset as follows:

```
if (!requireNamespace("remotes", quietly = TRUE)) {  
  install.packages("remotes")  
}  
# be sure to install from `2024` branch (code below does this)  
remotes::install_github("MiguelRodo/DataTidy23RodoHonsMult@2024")  
data("data_tidy_mali", package = "DataTidy23RodoHonsMult")
```

Question 1

Construct two-dimensional scatterplots of family versus distrd and of distrd versus cattle and remove any obvious outliers from the data set. (2)

Question 2

Comment on the need, or the lack thereof, for standardising the variables (i.e. dividing each by their standard deviation to yield unit variance). (1)

Question 3

Perform a principal component analysis using the correlation matrix using the function `eigen`. Determine the number of components to effectively summarize the variability. (5)

Question 4

Perform a PCA using the SVD, and confirm that the results match those using the correlation matrix. (2)

Question 5

Interpret the first two principal components using loadings. (2)

Question 6

Calculate correlations between the first principal component and individual variables, and compare these to the loadings. (2)

Question 7

Generate a biplot of the first 2 principal components based on standardised variables and interpret. (2)